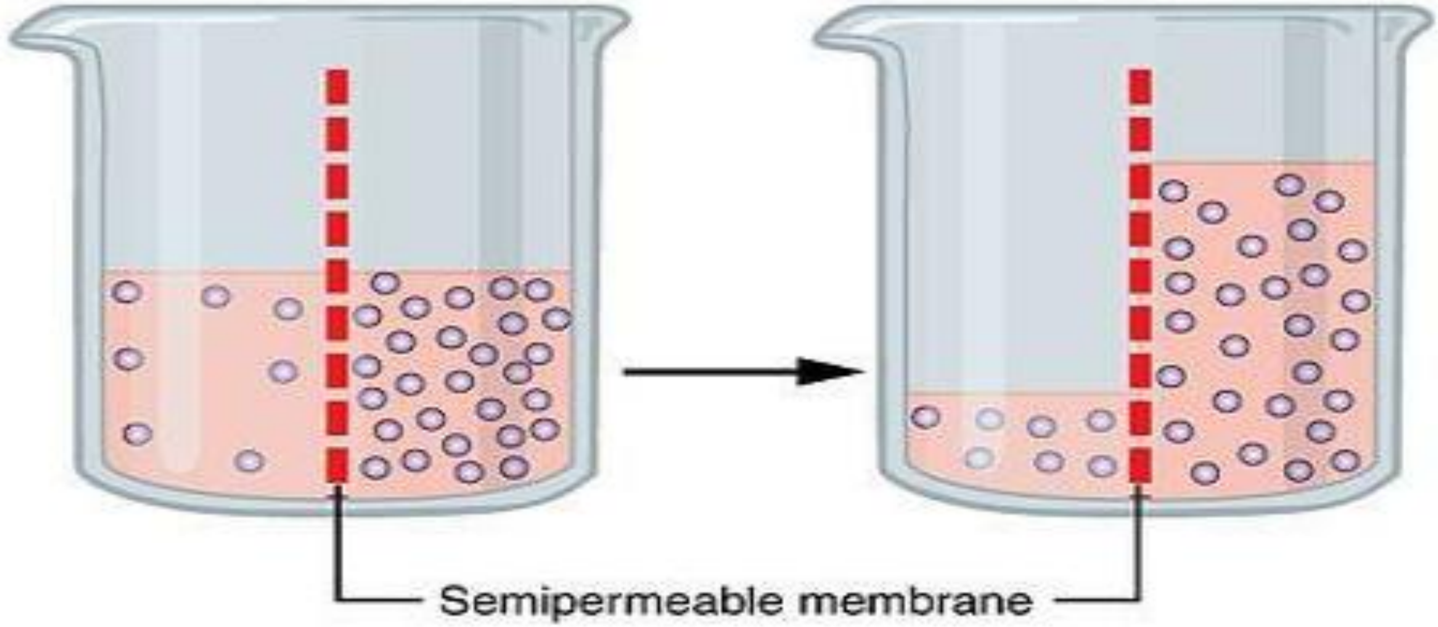
lec 13

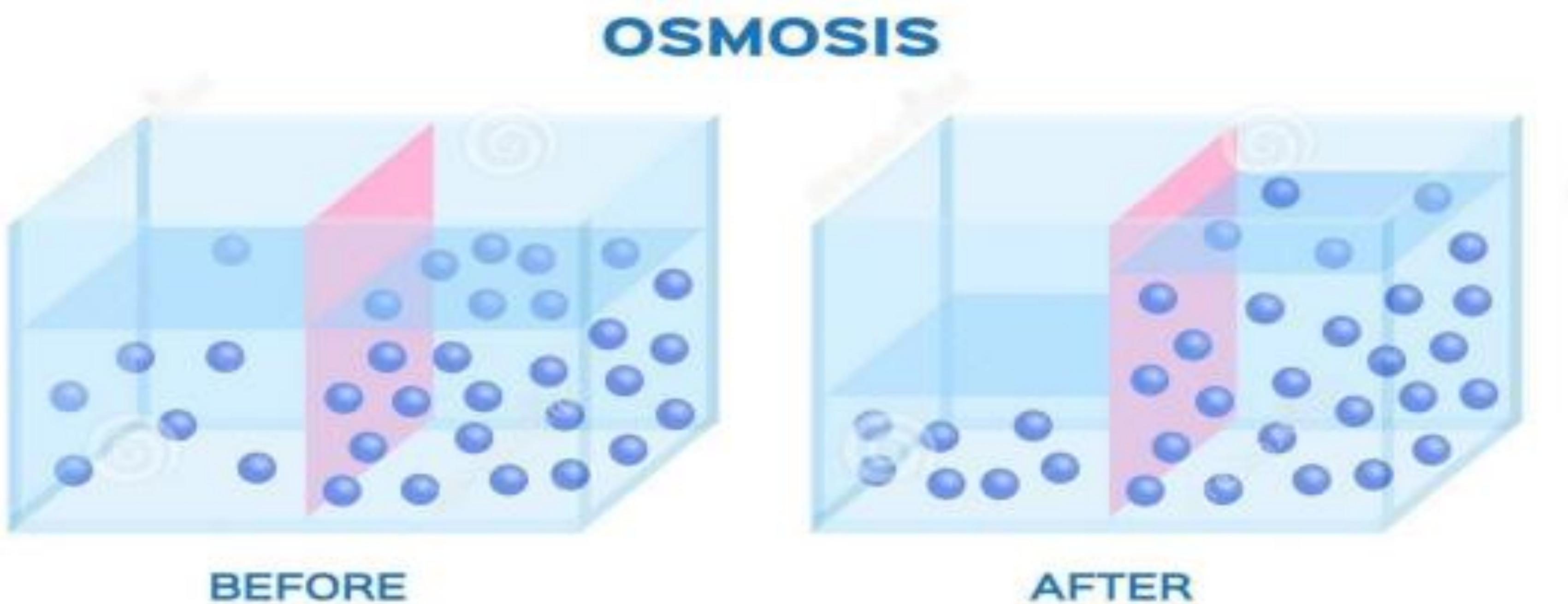
Methods of isotonicity adjustment

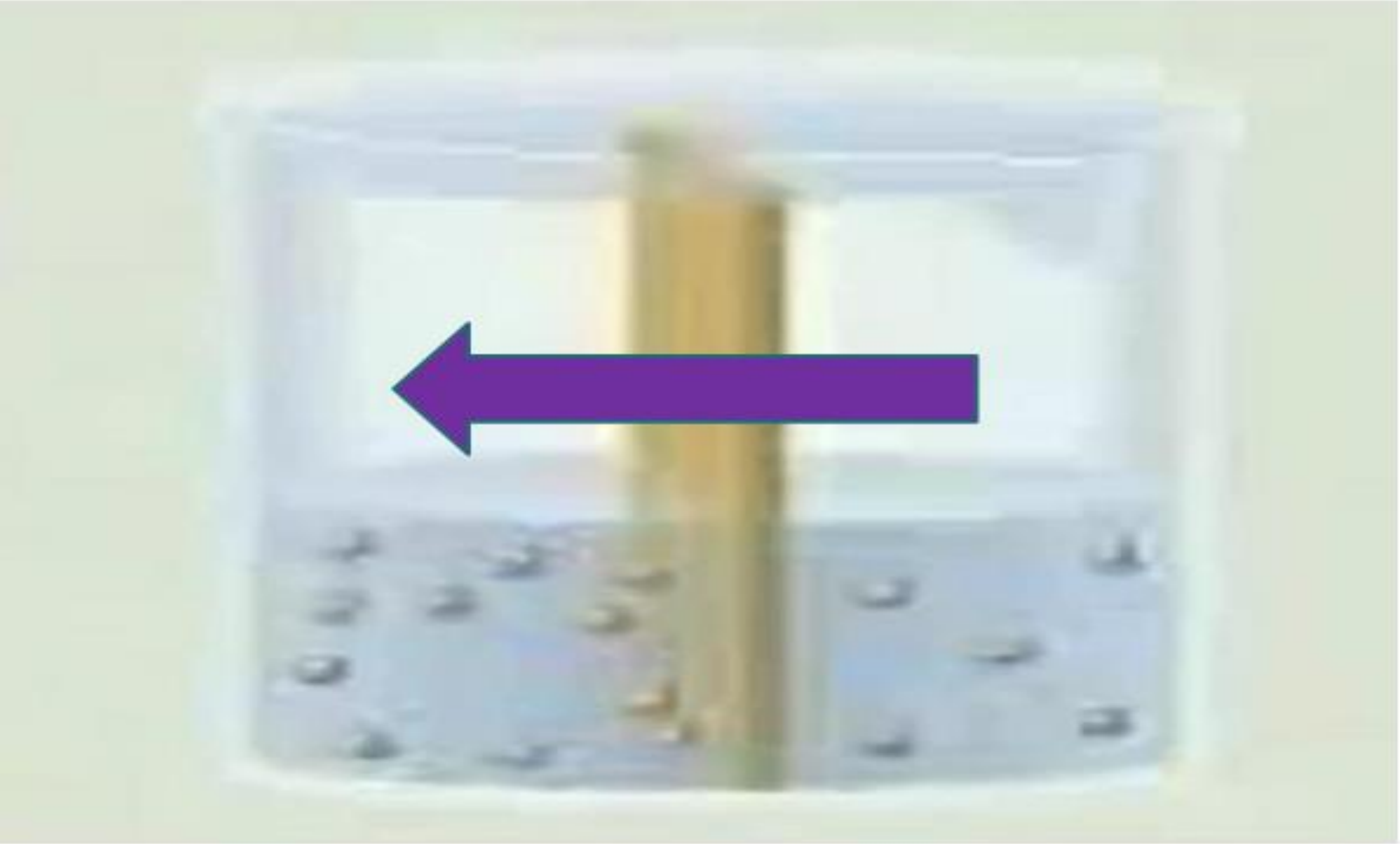
Buffer system

Osmotic pressure

- **Semipermeable membrane:** When in contact with a solution, they permit the passage of some molecules but not others
- They often permit the passage of small solvent molecules such as water but block the passage of larger solute molecules or ions
- This semipermeable character is due to a network of tiny pores within the membrane


Semipermeable membrane


OSMOSIS
BEFORE
AFTER



High solute
Low water

Low solute
High water

Osmosis & Osmotic Pressure

- Osmosis is the passage of a solvent across a semi-permeable membrane from an area of lower solute concentration (higher water) to an area of higher solute concentration (lower water) until equilibrium is reached.

- Thus, there is a net movement of solvent molecules from the less concentrated solution into the more concentrated one

- So, the amount of the liquid on the pure solvent side decrease with time and the amount of the liquid on the solution side increase with time and the concentration of the solution decrease

Isotonic solution: Two solutions of identical osmotic pressure are separated by a semipermeable membrane, no osmosis will occur as solutions have equal concentration of solute so equal osmotic pressure.

Hypertonic Solutions: Solution with higher concentration of solute and higher osmotic pressure, it is described hypertonic with respect to the dilute solution.

Hypotonic Solutions: one solution is of lower osmotic pressure, it is described as being hypotonic with respect to the more concentrated solution.

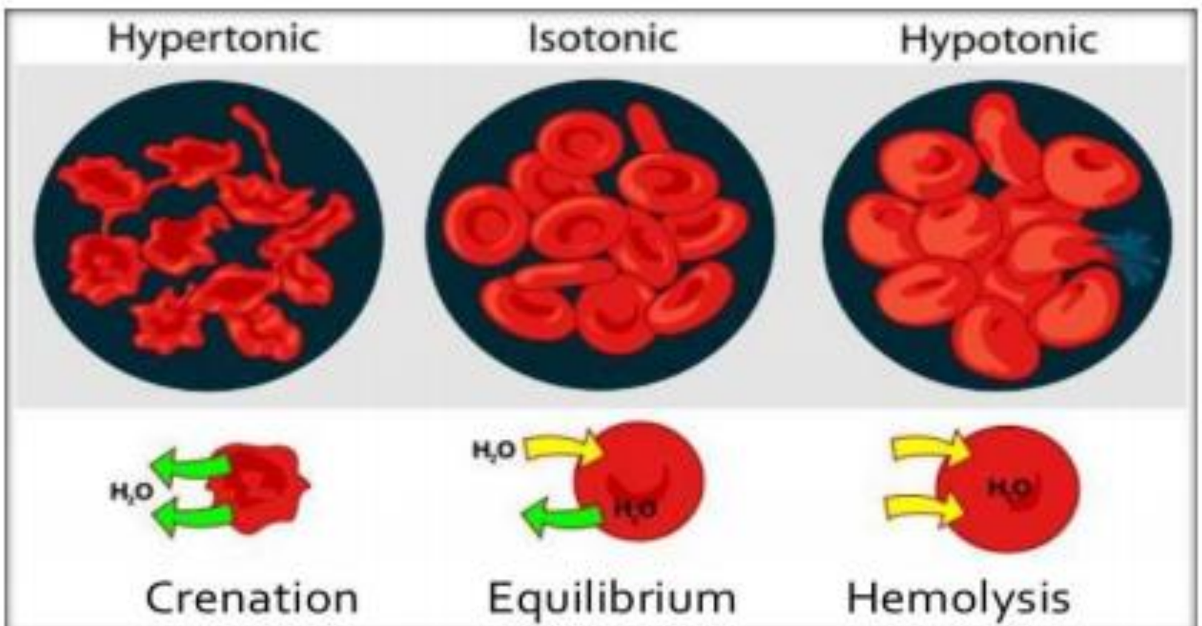
Hypotonic	Hypertonic	Isotonic
High water	Low water	Same water
Low solute	High solute	Same solute
Low osmotic pressure	High osmotic pressure	Same osmotic pressure
Gives water	Withdraws water	No exchange in water

Isotonic Solution

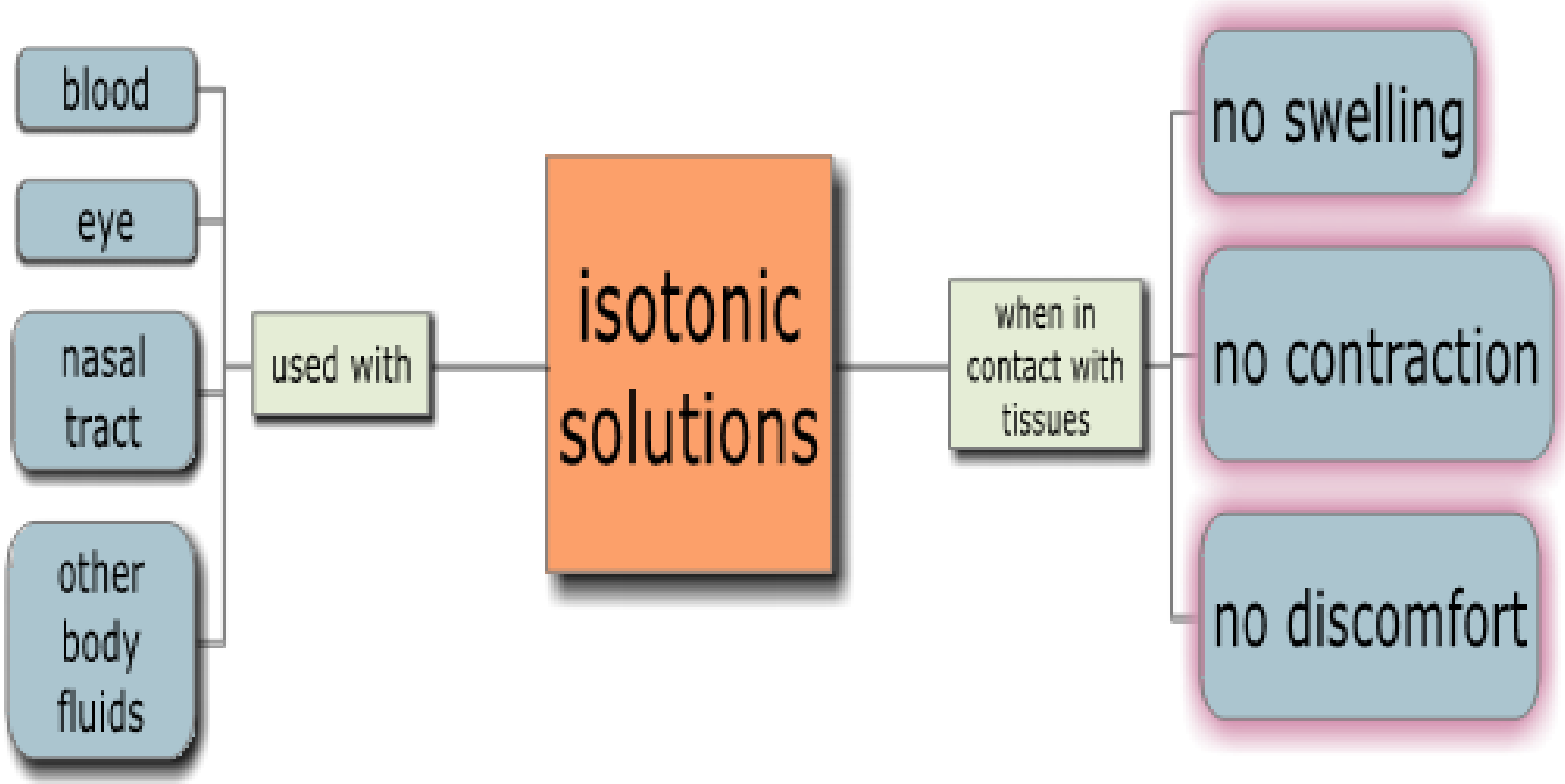
- Solutions to be mixed with body fluids should have the same osmotic pressure.

Higher osmotic pressure $\longrightarrow$ Hypertonic

Lower osmotic pressure $\longrightarrow$ Hypotonic



Why using isotonic solutions?



Methods of isotonicity adjustment:

- Freezing point depression method
- NaCl equivalent method
- White Vincent method
- Sprowls method

1) Freezing Point Depression

- Biological fluids such as serum and lachrymal fluids have a freezing point of -0.52 ˚c (Not 0 ˚c due to dissolved salts).

- So any solution with freezing point of -0.52 ˚c will have the same number of particles in solution and thus the same osmotic pressure. These solutions are called **Isotonic.**

- The most famous isotonic solution is normal saline (0.9% NaCl).

$$\frac{\text{Known percentage conc. of a given chemical}}{\text{Freezing point depression of the chemical at that concentration}} = \frac{X}{\text{Freezing point depression of blood plasma}}$$

Example 1:

1% NaCl solution has a freezing point depression of 0.576°C. What is the percentage concentration of NaCl required to make isotonic saline solution?

One can calculate the percentage concentration of NaCl by setting up the following proportion and solving for *X*:

$$\frac{1\%}{0.576} = \frac{X}{0.52}$$

$$X = (0.52 \times 1)/0.576$$

$$= 0.903 \text{ or } 0.9\% \text{ w/v}$$

What concentration of procain hydrochloride will yield a solution isotonic with blood plasma? Where the freeezing point of 1% w/v solution procain hydrochloride is – 0.122°C.

Answer:

$1\% \ / \ 0.122 = X / 0.52$

$X = 4.26$ % w/v

what concentration of cocaine hydrochloride will yield a solution iso-osmotic with blood plasma where the freezing point of 1% w/v solution of cocaine hydrochloride is -0.9 °C

Answer:

$1\% \ / \ 0.9 = X / 0.52$

$X = 1.73$ % w/v

NaCl equivalent method

- The sodium chloride equivalent of a drug is the amount of sodium chloride that has the same osmotic effect as 1 gm of the drug.
- NaCl equivalent (E):
  1g drug → Xg NaCl
- 0.9% NaCl solution is isotonic.
  0.9 gm NaCl / 100 ml solution

% of soldium chloride for adjustment =
0.9- (% strength of medicament x E)

A solution contains 1 gm of ephedrine sulfate in a volume of 100 ml, what quantity of sodium chloride must be added to make the solution isotonic ? How much dextrose would be required for this purpose? NaCl equivalent (E) of Ephedrine sulfate is 0.23 gm

Answer:1 gm of Ephedrine sulfate is osmotically equivalent to 0.23 gm NaCl

1 gm $\longrightarrow$ 0.23 gm

1 gm $\longrightarrow$??????

which is 1.0 gm X 0.23 = 0.23 gm

Since a total of 0.9 gm of sodium chloride is required for isotonicity of 100 ml

0.90 – (1.0 gm X 0.23) = 0.67 gm of NaCl must be added.

Dextrose

If it is desired to use dextrose instead of sodium chloride for adjusting the tonicity, the sodium chloride equivalent of dextrose = 0.16

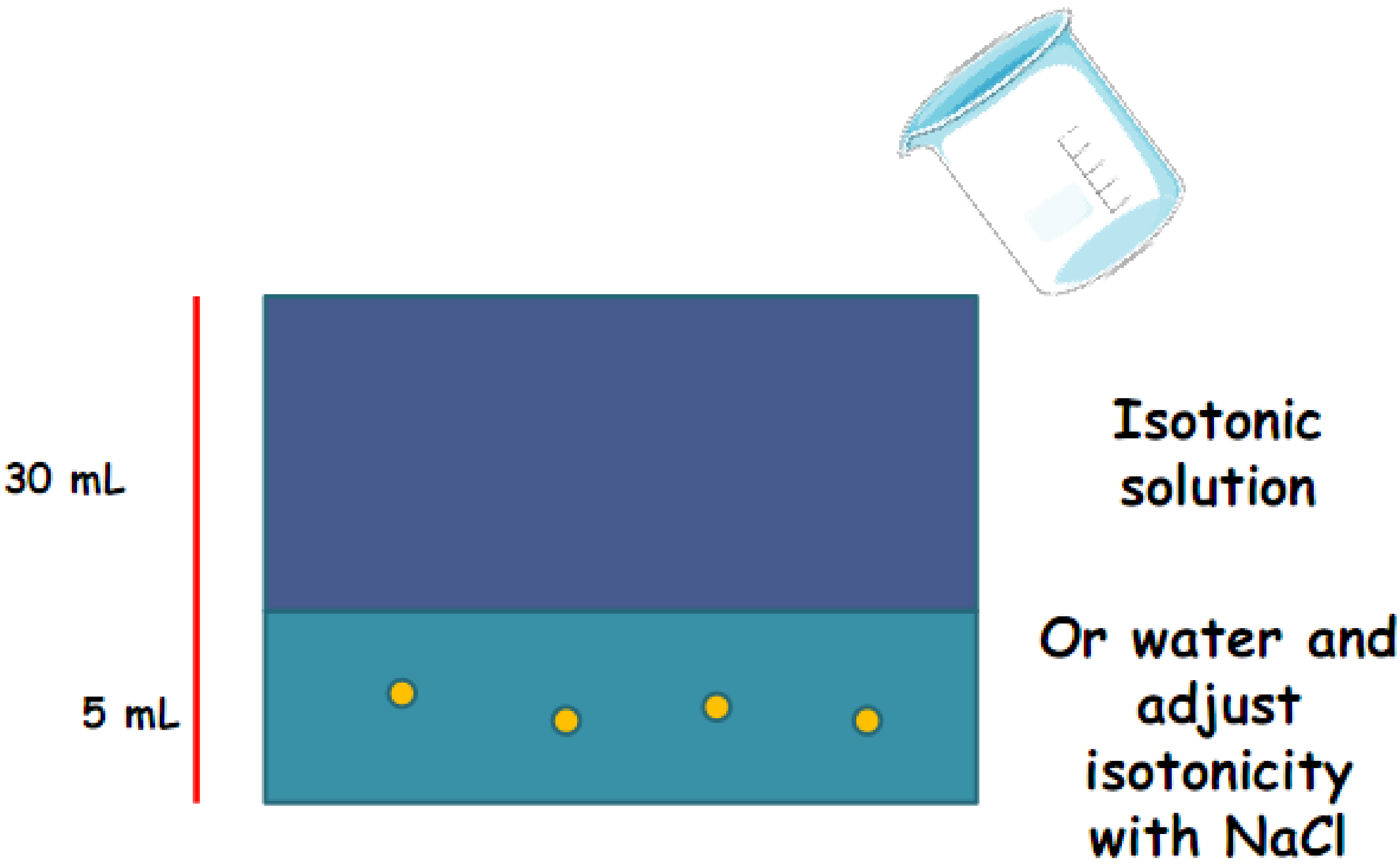
1gm dextrose = 0.16 gm NaCl

X gm dextrose = 0.67 gm NaCl

X = 4.2 gm of dextrose

White Vincent method

- This method depends on dissolving the amount of drug in a certain volume, to form isotonic solution, then adjusting to the final volume using already prepared isotonic solution.



White Vincent method

This method depends on dissolving the amount of drug in a certain volume, to form isotonic solution, then adjusting to the final volume using already prepared isotonic solution.

$$V = W \times E \times 111.1$$

V = Volume to be isotonic.

W = weight of drug added

E = NaCl equivalent of the drug

Example:

Rx

Phenacaine hydrochloride ---------------------------------0.06 g

Boric acid--0.3 g

Sterilized water to---100 ml

The sodium chloride equivalent of phenacaine hydrochloride is 0.16 and of boric acid is 0.5

Answer:

$V= W \times E \times 111.1$

$=[(0.06 \times 0.160 + (0.3\times 0.5)] \times 111.1$

$= 17.7$ ml

Buffers and Buffering Agents

Buffer solutions: are solutions of compounds or mixtures of compounds which resist changes in their pH upon addition of small quantities of an acid or alkali.

Composition: Most buffers solutions usually consist of a mixture of a weak acid and one of its salts or a weak base and one of its salts.

Buffer Action: It is the ability of buffer solutions to resist pH change upon addition of an acid or a base.

Example 1: NaCl solution in water → its pH 7 on addition of 1 ml 1N HCl solution (acid) to 1 liter of NaCl solution → pH from 7 well become about 3.

Example 2: On addition of 1 ml of 1N NaOH solution (alkali) to 1 liter of NaCl solution → raises pH from 7 to about 11.

From Example 1, 2: Sodium Chloride NaCl is not a buffer

Type of buffer solutions

1. **Acidic buffer solutions   2. Alkaline buffer solutions**

   1. **Acidic buffer solutions:**
      - ➢An acidic buffer solution is simply one which has a pH of less than 7.
      - ➢Acidic buffer solutions are commonly made from a weak acid and one of its (conjugate base) salt.

- Mixture of weak acid (acetic acid) and its salt (sodium acetate); Example: Acetate Buffer -mixture of acetic acid and sodium acetate in solution.

◊ The mixture thus consists of CH_3COOH molecules as well as $CH_3COO^- + Na^+$ ions.

If a strong acid is added to the mixture → H^+ ions supplied by the acid are immediately taken up by CH_3COO^- ions → slightly dissociated CH_3COOH

$$H^+ + CH_3COO^- \rightarrow CH_3COOH \text{ (slightly dissociated)}$$

The H^+ ions are neutralized by acetate ions → very little change in the pH of the mixture.

◊ On the other hand; if a strong base is added, the OH^- ions supplied by the base are neutralized by acetic acid present in the mixture → very little change in pH of solution. $OH^- + CH_3COOH \rightarrow CH_3COO^- + H_2O$

2. Alkaline buffer solutions:

An alkaline buffer solution has a pH greater than 7
Alkaline buffer solutions are commonly made from a weak base and one of its salts.

Example: mixture of ammonium hydroxide and largely dissociated salt ammonium chloride. Here the mixture contains the undissociated NH_4OH as well as $NH4^+$ + Cl^- ions

If a strong acid is added to the mixture → H^+ supplied by the acid is neutralized by the base NH_4OH $H^+ + NH_4OH \rightarrow H_2O + NH_4^+$

On the other hand, if a strong base is added, the OH^- ions are neutralized by NH_4^+ forming very slightly dissociated NH_4OH

$$OH^- + NH_4^+ \rightarrow NH_4OH \text{ (very slightly dissociated)}$$

From the previous two concepts, there is very little change in the pH

Common Pharmaceutical Buffers :

S.No	Name of Buffer	Component	pH
1	Acetate	Acetic acid + Sod.Acetate	4-5.6
2	Citrate	Citric acid + Sod.citrate	3-6.2
3	Borate	Boric acid + Sod.borate	7.4-9.2
4	Phosphate	DiSodium mono hydrogen phosphate + Mono Sod.Di hydrogen Phosphate	5.8-8

Factors Affecting pH of buffer Solutions

1. **Temperature:** Change in temperature → changes Buffer pH with Buffers consisting of a base and its salt show greater changes with temperature

↑ in temperature → ↓ the pH of a basic buffer solution

On the other hand, ↑ in temperature → ↑ the pH of a buffer solution containing acetic acid and sodium acetate

2. Dilution effect: Dilution with water in moderate quantities small effect on the pH of the buffer

Dilution of acidic buffer → ↑ in pH

Dilution of basic buffer → ↓ in pH

3. Salts effect: When a salt is added to an acidic buffer solution → ↓ pH

When a salt is added to a basic buffer → ↑ its pH

Biological Buffers

Blood pH of blood is maintained at 7.4 by the component of plasma and erythrocytes. The plasma contains:

1- Carbonic acid/carbonate 2- Phosphoric acid / sodium phosphate

Erythrocytes (RBSs) contains:

1- Oxyhaemoglobin/haemoglobin 2- Phosphoric acid / potassium phosphate

The pH range of blood is 7.35 → 7.45 (above or below this range → life in danger)

Lachrymal fluid or tears: pH is about 7.4 (range 7 → 7.8)

Urine : Is about 6 (range 4.5 → 7.8) when the pH of urine is altered above or below this range kidney is diseased.

Pharmaceutical Buffers

In pharmaceutical industry we adjust the pH of the product for maximum stability or maintain the pH of the product within the optimum physiological range.

1- Buffers in tablet formulation: Buffers employed in tablet or capsule formulations containing acidic drugs to reduce gastric irritation or antacid drugs such as sodium bicarbonate, sodium citrate or magnesium carbonate.

2- Buffers in ophthalmic preparations: Buffers are used here to maintain the pH of the product within the pH range of lachrymal fluids. The lachrymal fluid has good buffering capacity where the solutions within 3.5 – 10.5 can usually be tolerated Outside this range → eye irritation and increase lacrimation. The most commonly used buffers in ophthalmic preparations include borate, phosphate and carbonate buffers.

3- Buffers in parenteral preparations: In parenteral, the consideration of pH is very important, where high pH (above 9) → tissue necrosis while highly acidic pH (below 3) extreme pain in site of injection.

The ideal pH for parental is 7.4 similar to the blood. There must be compromised between the blood pH and the solubility and stability of the product. The most widely used buffers in parental are acetate, phosphate, citrate and glutamate buffers.

4- Buffers in creams and Ointments: Topical preparations on storage usually render change in pH, this affect the stability of the drugs therefore, buffers must be induced in this formulations. The most widely used buffers here are citrate and phosphate.

Buffered Isotonic Solutions

Iso osmotic or isotonic solutions, if they exert the same osmotic pressure when separated by a semi-permeable membrane. Definition from the physiological view; isotonic solutions are solutions having the some osmotic pressure as that of the body fluids when separated by a biological membrane. Body fluids (blood and lachrymal fluids) have osmotic pressure corresponding to that of 0.9 % solution sodium chloride. Thus 0.9 % solution sodium chloride is called isotonic with physiological fluids.

- Solutions with osmotic pressure low than 0.9% NaCl → hypotonic solution
- Solutions with osmotic pressure higher than 0.9% NaCl → hypertonic solution

Thank You